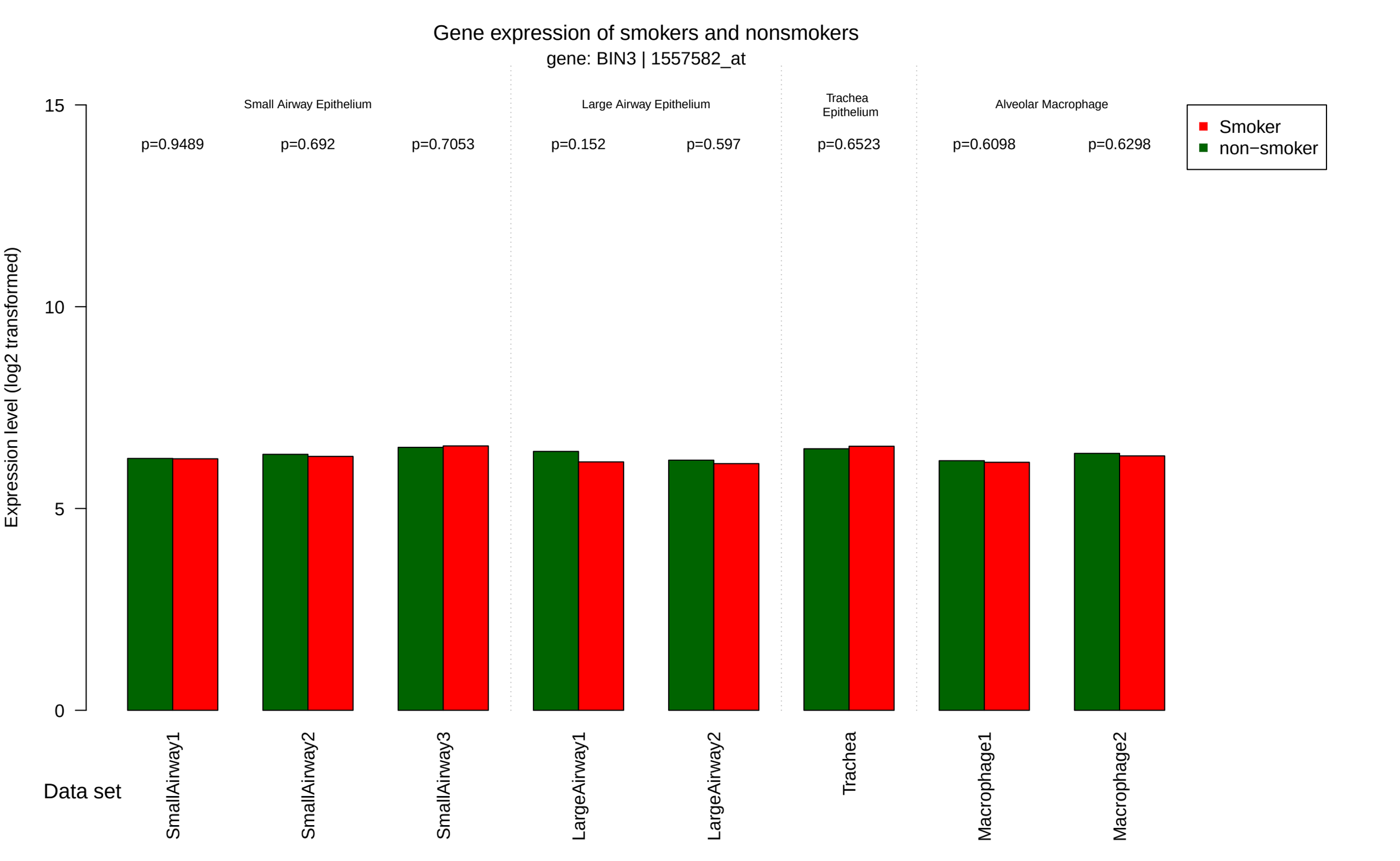
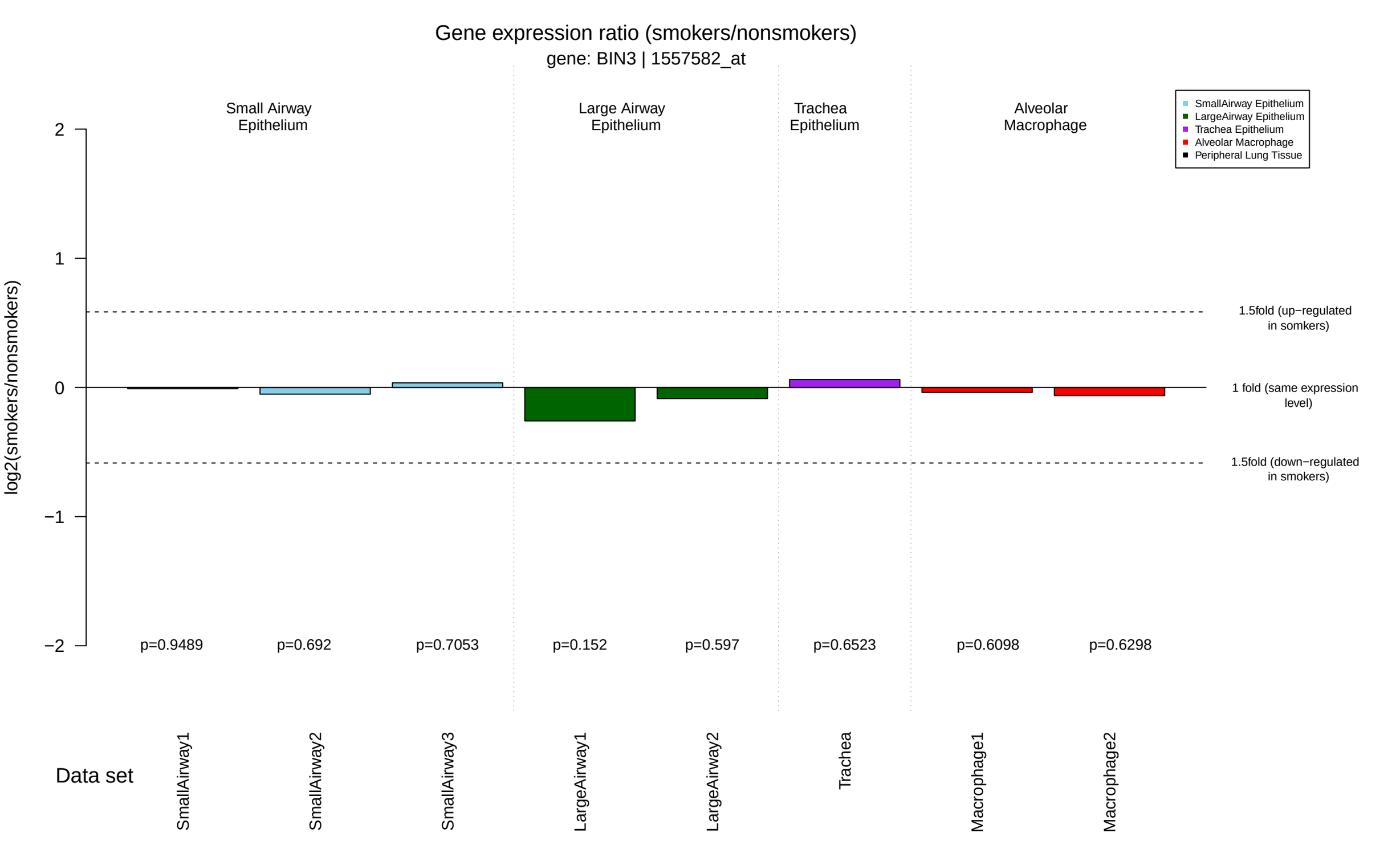
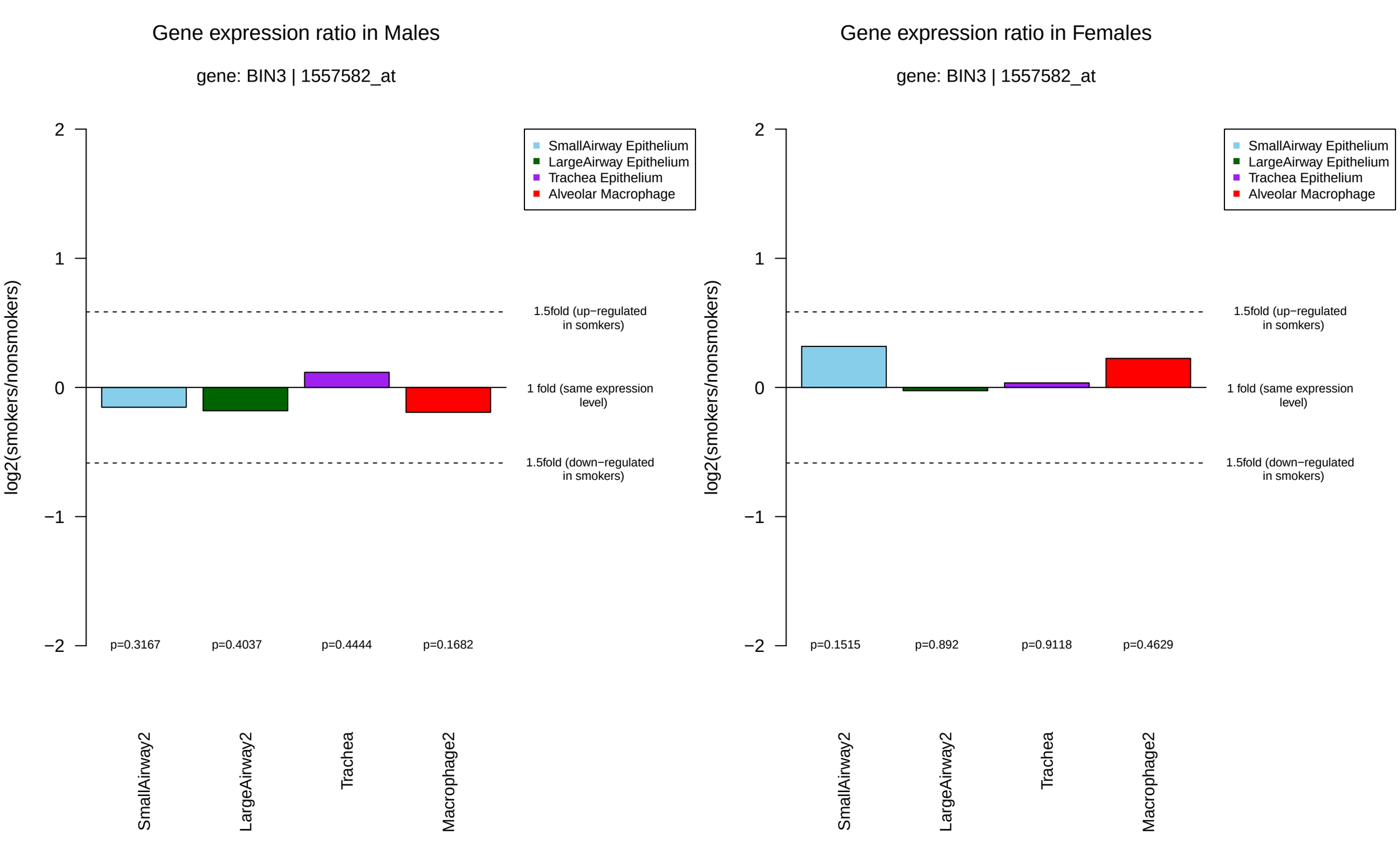
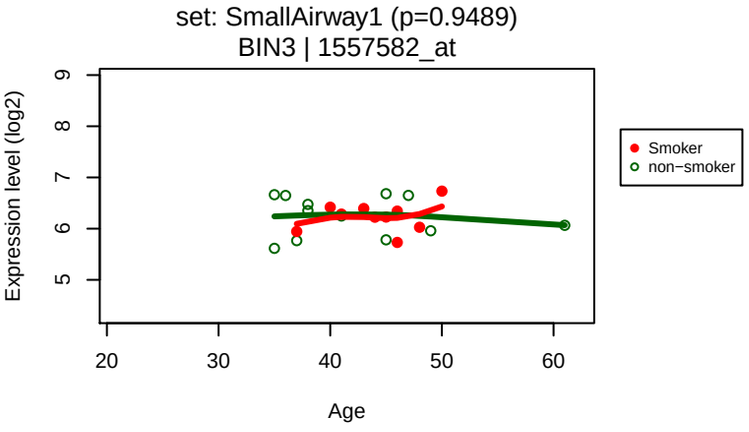
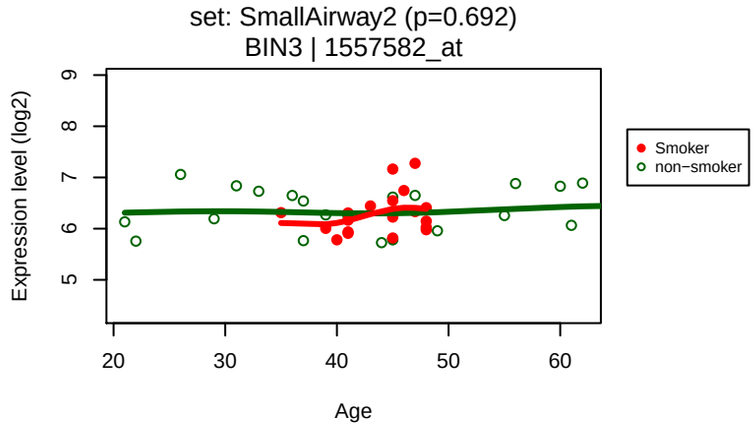
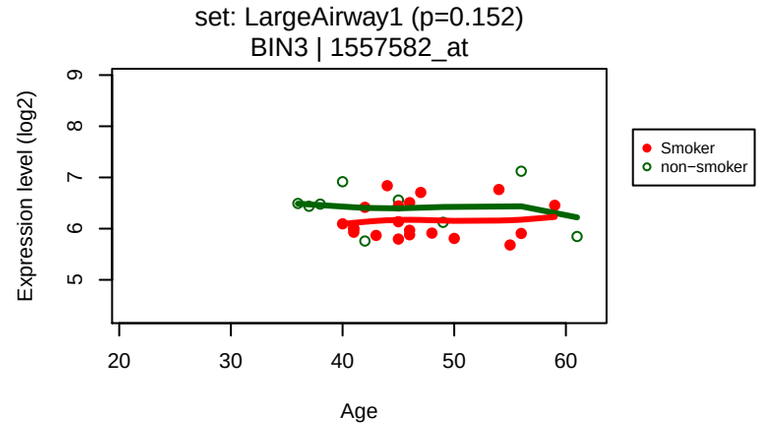

Gene expression of smokers and nonsmokers
gene: BIN3 | 1557582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9489
p=0.692
p=0.7053
p=0.152
p=0.597
p=0.6523
p=0.6098
p=0.6298
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

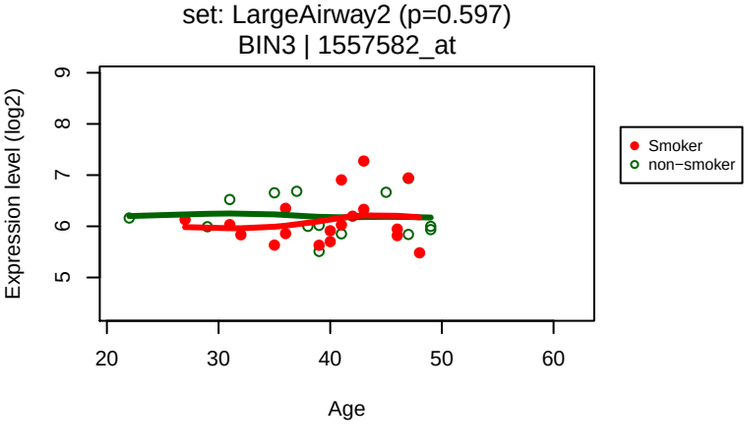

Gene expression ratio (smokers/nonsmokers)
gene: BIN3 | 1557582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9489
p=0.692
p=0.7053
p=0.152
p=0.597
p=0.6523
p=0.6098
p=0.6298
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

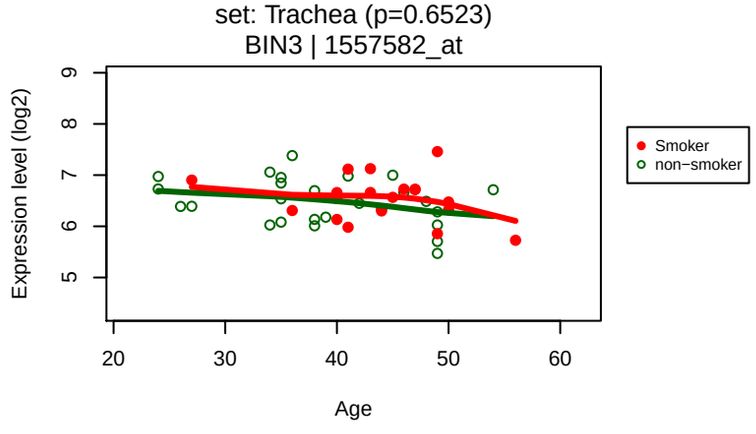

Gene expression ratio in Males
gene: BIN3 | 1557582_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3167
p=0.4037
p=0.4444
p=0.1682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BIN3 | 1557582_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1515
p=0.892
p=0.9118
p=0.4629
SmallAirway2
LargeAirway2
Trachea
Macrophage2

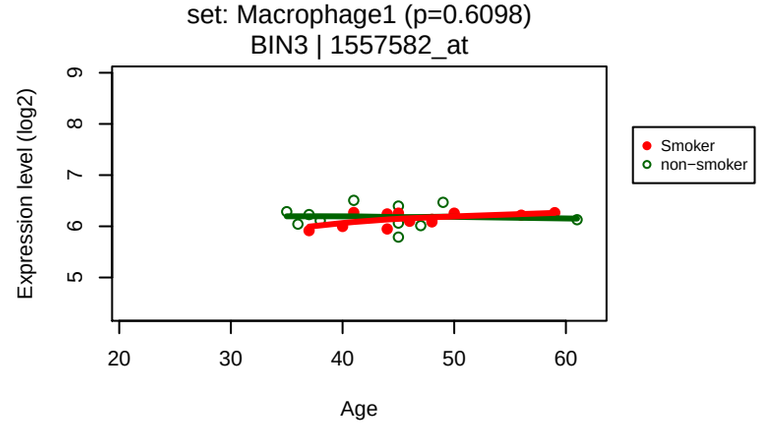

set: SmallAirway1 (p=0.9489)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker

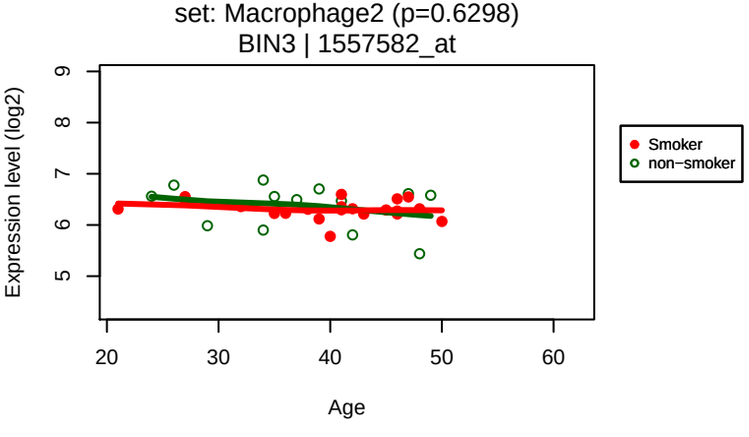

set: SmallAirway2 (p=0.692)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker

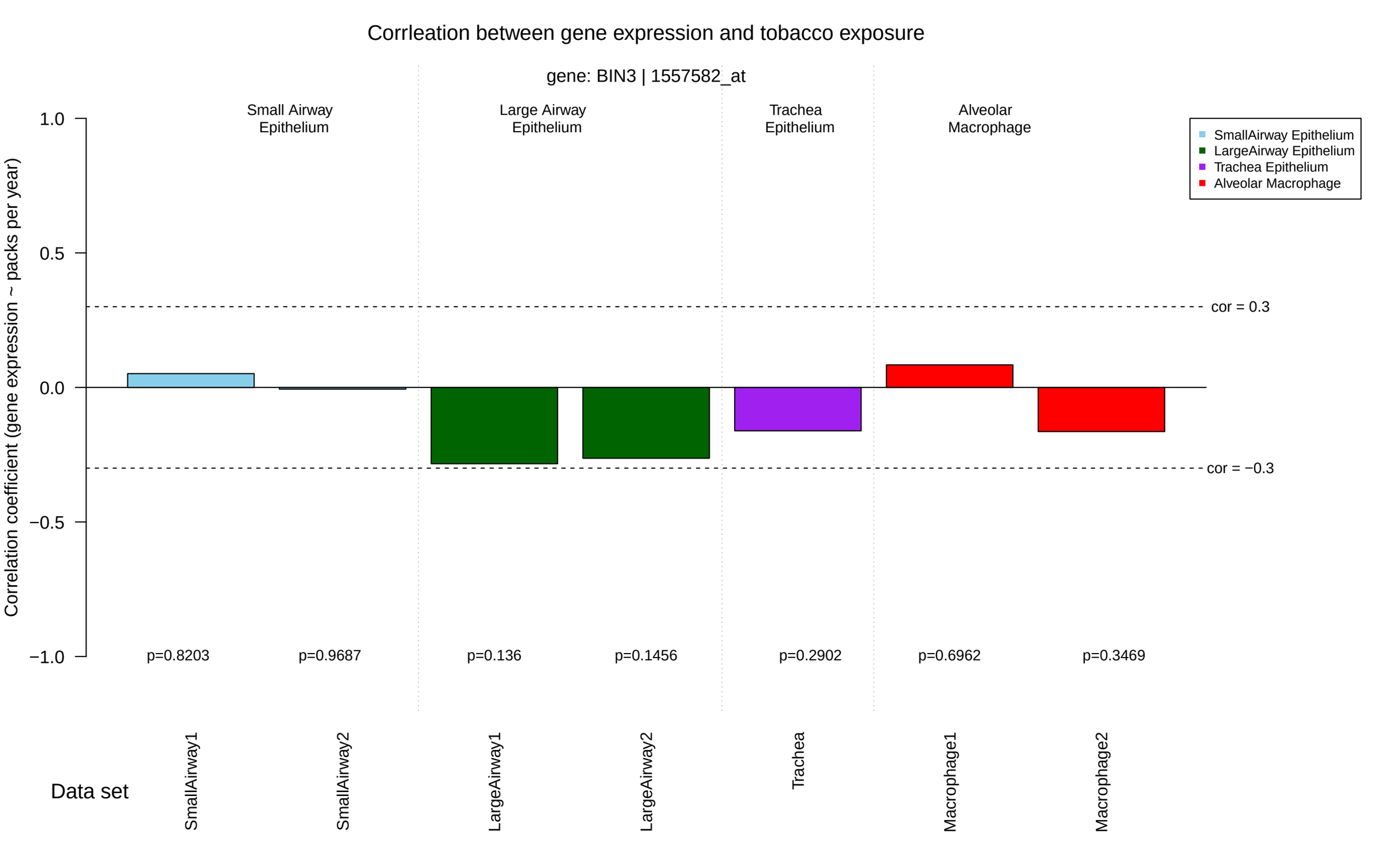

set: LargeAirway1 (p=0.152)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.597)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6523)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6098)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6298)
BIN3 | 1557582_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BIN3 | 1557582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8203
p=0.9687
p=0.136
p=0.1456
p=0.2902
p=0.6962
p=0.3469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2